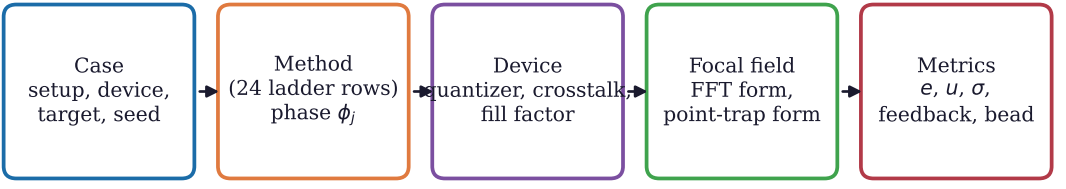

the same chain runs offline in Python (reference) and live in the browser (TypeScript port)